

$$\begin{array}{ccc}
 G(M, E) & \xrightarrow{\eta_{G(M, E)}^R} & RG(M, E) \\
 & \searrow & \downarrow \chi_E^M \\
 & & G(M, TE)
 \end{array}$$

$G(M, \eta_E^T)$